February 14, 2025

1. If X is a discrete random variable having the PMF

x	1	2	3	4
P(x)	0.4	0.3	0.2	0.1

Find $P(\frac{1}{2} < X < \frac{7}{2}/X > 1)$.

2. If $P(X = x) = \begin{cases} kx & x = 1, 2, 3, 4, 5 \\ 0 & \text{otherwise} \end{cases}$ represents a probability, find

   (a) the value of k,

   (b) P(X being a prime number),

   (c) Find $P(\frac{1}{2} < X < \frac{5}{2}/X > 1)$, and

   (d) the distribution function of X.

3. A random variable X has the following probability function:

x	0	1	2	3	4	5	6	7
P(x)	0	k	2k	2k	3k	k^2	$2k^2$	$7k^2 + k$

   (a) Find k,

   (b) Evaluate $P(X < 6)$, $P(X \geq 6)$ and $P(0 < X < 5)$,

   (c) Find the minimum value of m for which $P(X \leq m) > \frac{1}{2}$,

   (d) Mean of X, and

   (e) the distribution function of X.

4. If X is a continuous random variable whose PDF is given by

$$f(x) = \begin{cases} k(4x - 2x^2), & 0 < x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

Find

   (a) the value of k,

(b) $P(X > 1)$.

5. A random variable X has the PDF given by

$$f(x) = \begin{cases} cxe^{-x}, & x > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Find

(a) the value of c,

(b) the CDF.

6. If X is a continuous random variable with PDF

$$f(x) = \begin{cases} x, & 0 \le x < 1, \\ \frac{3}{2}(x-1)^2, & 1 \le x \le 2 \\ 0, & \text{otherwise.} \end{cases}$$

Find

(a) the CDF,

(b) $P\left(\frac{3}{2} < x < \frac{5}{2}\right)$.

7. If a random variable X has the probability density function

$$f(x) = \begin{cases} \frac{1}{2}(x+1), & -1 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find the mean and variance.

8. Find the MGF of the random variable X whose probability function $P(X = x) = \frac{1}{2^x}$, $x = 1, 2, 3 \cdots$. Hence find its mean.

9. A random variable X has the PDF given by

$$f(x) = \begin{cases} 2e^{-2x}, & x \ge 0, \\ 0, & \text{otherwise.} \end{cases}$$

Find

(a) MGF of X,

(b) the first two moments about the origin.

10. The mean and variance of a binomial distribution are 16 and 8. Find $P(X \ge 3)$.

11. Four coins were tossed simultaneously. What is the probability of getting

(a) 3 heads,

(b) at least 3 heads,

(c) at most 3 heads.

12. Find the probability that in tossing a fair coins 5 times. there will appear

    (a) 3 heads,

    (b) 3 heads and 2 tails,

    (c) at least 1 head and

    (d) not more tha 1 head.

13. Sixteen coins are thrown simultaneously. Find the probability of getting at least 7 heads.

14. In 256 sets of 12 tosses of a fair coin, in how many cases may one expect 8 heads and 4 tails?

15. If X is a Poisson random variable with $P(X = 1) = P(X = 2)$, find $P(X \geq 3)$.

16. The number of monthly breakdown of a computer is a random variable having a Poisson distribution with mean equal to 1.8. Find the probability, that this computer will function for a month with

    (a) only 1 breakdown,

    (b) at least 1 breakdown.

17. The atoms of a radioactive element are randomly disintegrating. If every gram of this element, on average, emits 3.9 alpha particles per second, what is the probability that during the next second the number of alpha particles emitted from 1g is

    (a) at most 6,

    (b) at least 2,

    (c) at least 3 and at most 6.

18. A car hire firm has 2 cars. The number of demands for a car on each day is distributed as Poisson variate with mean 0.5. Calculate the proportion of days on which

    (a) neither car is used,

    (b) some demands is refused.

19. If X is a Poisson variate such that $P(X = 1) = \frac{3}{10}$ and $P(X = 2) = \frac{1}{5}$, find $P(X = 0)$ and $P(X = 3)$.

20. The average number of phone calls/minute coming into a switch board between 2pm and 4pm is 2.5. Determine the probability that during one particular minute there will be

    (a) 4 or fewer,

    (b) more than 6 calls.

(Assuming that number of phone calls is Poisson random variable!)

21. Fit a Poisson distribution to the following data and calculate the theoretical frequencies

Deaths	0	1	2	3	4
Frequencies	122	60	15	2	1

22. Let X be a normal variable with mean μ and standard deviation σ. If Z is the standard normal variable such that $Z = -0.8$ when $X = 26$ and $Z = 2$ when $X = 40$. Then find μ and σ and also find $P(X > 45)$ and $P(|X - 30| > 5)$.

23. If $Z \sim N(0, 1)$, find the value of the constant c such that $P(|Z| \leq c) = 0.95$.

24. Find the mean and standard deviation of a normal distribution in which 7% of the items are under 35 and 89% are under 63.

25. A sales tax officer has reported that the average sales of the 500 businesses that he has to deal with during a year is Rs. 36,000 with a standard deviation of Rs. 10,000. Assuming that the sales in these businesses are normally distributed, find

    (a) the number of businesses as the sales of which is above Rs. 40,000,

    (b) the percentage of businesses as the sales of which are likely to range between Rs. 30,000 and Rs. 40,000.

26. In a test on electric bulbs, it was found that the life of a particular brand was distributed normally with an average life of 2000 hours and SD of 60 hours. If a firm purchases 2500 bulbs find the number of bulbs that are likely to last for

    (a) more than 2100 hours,

    (b) less than 1950 hours,

    (c) between 1900 to 2100 hours.